HW 7

Pr. 1.

Change of variables. Suppose we have the PDE

(1) $$3u_x - 2y u_y + 3u = 0$$

and we want to change independent vars from (x, y) to (ξ, η):

(2) $$\xi = 2x + y$$

(3) $$\eta = 3x + 2y$$

Show that the PDE in terms of ξ & η is (careful - Prof. Lynn may have made a mistake)

(4) $$(6 + 6\xi - 4\eta) u_\xi + (9 + 12\xi - 8\eta) \cdot u_\eta + 3u = 0.$$

Sol-n:

To have the given change of variables, we should present u_x, u_y's in terms of ξ & η:

Note that

(5) $$u = u(\xi, \eta) = u(\xi(x,y), \eta(x,y)), \quad \text{hence}$$

(6) $$u_x = \frac{\partial u}{\partial \xi} \cdot \left.\frac{\partial \xi}{\partial x}\right|_y + \frac{\partial u}{\partial \eta} \cdot \left.\frac{\partial \eta}{\partial x}\right|_y = u_\xi \cdot \xi_x + u_\eta \cdot \eta_x \quad \text{&}$$

(7) $$u_y = u_\xi \cdot \xi_y + u_\eta \cdot \eta_y, \quad \text{also}$$

(8) $$\begin{cases} \xi = 2x + y \\ \eta = 3x + 2y \end{cases} \qquad \begin{bmatrix} \xi \\ \eta \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}}_{J} \begin{bmatrix} x \\ y \end{bmatrix}$$

$\det(J) = 1 \neq 0 \Rightarrow$ the transformation is valid. so $\Rightarrow$

(9) $$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}^{-1} \cdot \begin{bmatrix} \xi \\ \eta \end{bmatrix} = \frac{1}{1} \cdot \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \cdot \begin{bmatrix} \xi \\ \eta \end{bmatrix} \Rightarrow \begin{cases} x = 2\xi - \eta \\ y = -3\xi + 2\eta \end{cases}.$$

So $\xi = 2x + y \qquad \eta = 3x + 2y$

$$\begin{cases} \xi_x = 2, & \eta_x = 3 \\ \xi_y = 1 & \eta_y = 2 \end{cases}$$

plugging these into (6) & (7) results

$$u_x = u_\xi \cdot (2) + u_\eta \cdot (+3) = 2u_\xi + 3u_\eta$$

$$u_y = u_\xi (1) + u_\eta \cdot (2) = u_\xi + 2u_\eta$$

So (1) becomes

$$3u_x - 2y\, u_y + 3u = 3 \cdot (2u_\xi + 3u_\eta) - 2 \cdot \overbrace{(-3\xi + 2\eta)}^{y} \cdot (u_\xi + 2u_\eta) + 3u$$

$$= 6u_\xi + 9u_\eta + (6\xi - 4\eta) \cdot (u_\xi + 2u_\eta) + 3u =$$

$$= 6u_\xi + 9u_\eta + 6\xi \cdot u_\xi + 12\xi u_\eta - 4\eta u_\xi - 8\eta u_\eta + 3u =$$

(10) $(6 + 6\xi - 4\eta) \cdot u_\xi + (12\xi + 9 - 8\eta) \cdot u_\eta + 3u = 0.$ which is of course not different from

(4) $(6 + 6\xi - 4\eta) \cdot u_\xi + (12\xi + 9 - 8\eta) u_\eta + 3u = 0.$

Confirming the work of Professor is true!